



CMU-CS 462: Software Measurement and Analysis

Spring 2025-2026

Nguyen Duc Man

E: mannd@duytan.edu.vn

T: 0904235945

12/01/2026

1

1



The basics of measurement

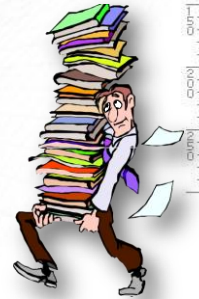
12/01/2026

2

2

Contents

1. Metrology (the scientific study of measurement)
2. Property-oriented measurement
3. Meaningfulness in measurement
4. Scale
5. Measurement validation
6. Object-oriented measurement



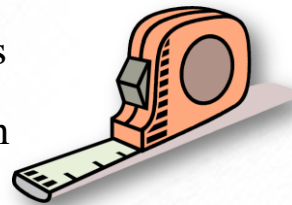
12/01/2026

3

3

4. Scale: Informal Definition

- A *measurement scale* is a set of predefined symbols or values in order to represent certain common measures.
- A *scale* is an abstract measurement tool for measuring defined common attributes of entities.



12/01/2026

4

4

Scale: Formal Definition

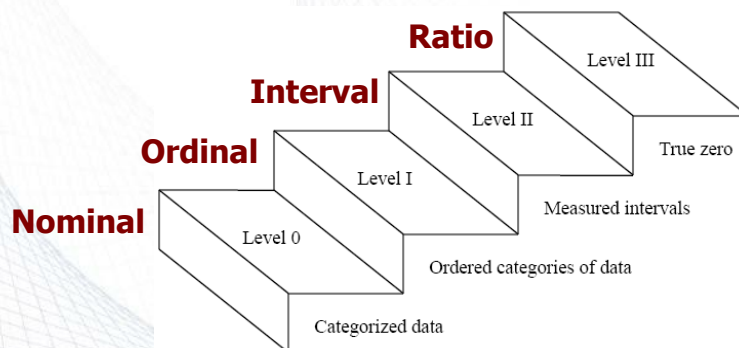
- Assume that the followings are given:
 - $A = \{a, b, c, \dots, z\}$ set of real world entities
 - k : property for each element of A
 - $\underline{A} = \{a, b, c, \dots, z\}$ is the model of A which describes each element of A in terms of the property k
 - The empirical and formal relational systems E and F
 - m , which is a mapping $\underline{A} \rightarrow \underline{A}'$
- Then $S = (E, F, m)$ is called the **scale** of measurement for the property k of the target set A
- If F is defined over a subset of real numbers, within the scale, the measurement maps the key property of each object of the target set into a number.

12/01/2026

5

5

Scales (Summary)



12/01/2026

6

6

Measurement Scale: Questions

- How do we know if a scale appropriately represents the relationships between measured properties of entities? (representation problem)



What do we do when we have several different scales for the same measure? (uniqueness problem)

- For a defined scale, what arithmetic operations make sense for measurement values?

12/01/2026

7

7

Scale Types

- Objective (regular) scales
 - Nominal, ordinal, interval, ratio, absolute
- Subjective scales
 - Likert-Type scale (Evaluation-Type, Frequency-Type, Agreement-Type)
 - Semantic differential scale
 - Summative scale

12/01/2026

8

8

Regular Scales

- The scale (E, F, m) is regular if and only if
 - for every other scale (E, F, g) , and
 - for all a and b in A ,

$$m(a) = m(b) \quad \text{implies} \quad g(a) = g(b)$$
- This is a rephrase for uniqueness property along various scales.
- If two objects measured on one scale yield to the same measure for the property k , the same holds on any other scale.

12/01/2026

9

9

Objective Scale: Types /1

Nominal Scales

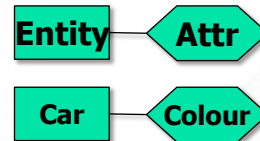
- Define classes or categories, and then place each entity in a particular class or category, based on the value of the attribute.
- **Properties:**
 - The empirical relation system consists only of different classes; there is **no notion of ordering** among the classes.
 - Any distinct numbering or symbolic representation of the classes is an acceptable measure, but there is **no notion of magnitude** associated with the numbers or symbols.

12/01/2026

10

Example: Nominal Scale

- Classification of cars based on their colour
- Classification of faults in a software:
 - Specification fault
 - Design fault
 - Coding fault



C-C1	1	White
C-C2	2	Yellow
C-C3	3	Red
C-C4	4	Blue
C-C5	5	Green
C-C6	6	other

Measure (Car Colour) $\in \{ "1", "2", "3", "4", "5", "6" \}$
 $\{ \text{White, ..., other} \}$

12/01/2026

11

11

Objective Scale: Types /2

Ordinal Scales

- The ordinal scale is useful to augment the nominal scale with information about an ordering of the classes or categories.
- **Properties:**
 - The empirical relation system consists of classes that are ordered with respect to the attribute.
 - Any mapping that **preserves the ordering** (that is, any monotonic function) is acceptable.
 - The numbers represent ranking only, so addition, subtraction, and other arithmetic operations have no meaning.

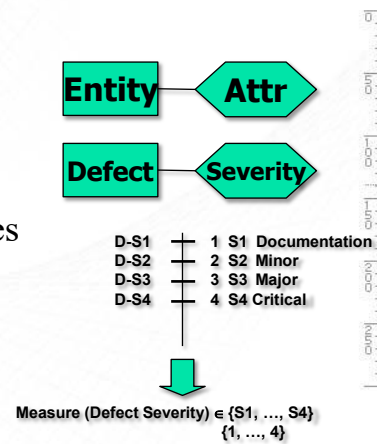
12/01/2026

12

12

Example: Ordinal Scale

- $M(x) > M(y)$ then $M'(x) > M'(y)$
- Measuring complexity of many software modules by defining 5 complexity classes
 - Trivial
 - Simple
 - Moderate
 - Complex
 - Incomprehensible



12/01/2026

13

13

Objective Scale: Types /3

Interval Scales

- Interval scale carries more information than ordinal and nominal scale. It captures information about the size of the intervals that separate the classes, so that we can in some sense understand the size of the jump from one class to another.
- **Properties:**
 - An interval scale **preserves order**, as with an ordinal scale.
 - An interval scale **preserves differences but not ratios**. That is, we know the difference between any two of the ordered classes in the range of the mapping, but computing the ratio of two classes in the range does not make sense.
 - Addition and subtraction are acceptable on the interval scale, but not multiplication and division.

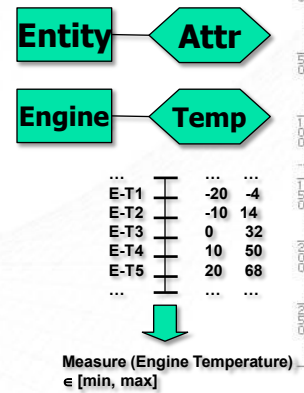
12/01/2026

14

14

Example: Interval Scale

- $M' = aM + b$
- Temperature ranges in Celsius and Fahrenheit
- Project scheduling
 - Requirement analysis 3 weeks
 - Design 4 weeks
 - Coding 4 weeks
 - Testing starts after coding is done
 - When testing starts? After 11 weeks



12/01/2026

15

15

Objective Scale: Types /4

Ratio Scales

- Sometimes we say that one liquid is twice as hot as another, or that one project took twice as long as another. This needs the ratio scale, which is the most useful scale of measurement.
- **Properties:**
 - It is a measurement mapping that **preserves ordering**, **preserves size of intervals** between entities, and **preserves ratios** between entities.
 - There is a zero element, representing total lack of the attribute.
 - The measurement mapping must start at zero and increase at equal intervals, known as units.
 - All arithmetic can be meaningfully applied to the classes in the range of the mapping.

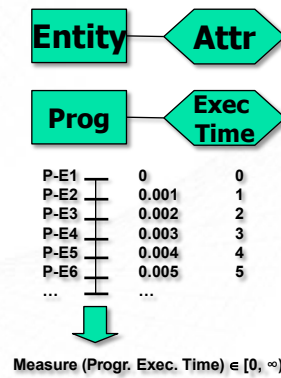
12/01/2026

16

16

Example: Ratio Scale

- Measuring length, distance, etc. preserves $M' = aM$
- Measuring execution time of a program



12/01/2026

17

17

Objective Scale: Types /5

Absolute Scales

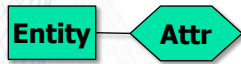
- The absolute scale is the most restrictive of all. For any two measures, M and M' , there is only one admissible transformation: the identity transformation.
- **Properties:**
 - The measurement for an absolute scale is made simply by counting the number of elements in the entity set.
 - The attribute always takes the form “number of occurrences of x in the entity.”
 - There is only one possible measurement mapping.
 - All arithmetic analysis of the resulting count is meaningful.

12/01/2026

18

18

Scale Types (Summary)



Measure (Attribute) is well-defined, if scale and unit are clearly specified; specification of the unit makes the measure unambiguous!

- **Nominal scale:** classification of objects, where the fact that objects are different is preserved
- **Ordinal scale:** objects are ranked/ordered according to some criteria, but no information about the distance between the values is given
- **Interval scale:** differences between values are meaningful
- **Ratio scale:** there is a meaningful “zero” value, ratios between values are meaningful
- **Absolute scale:** no transformation (other than identity) is meaningful

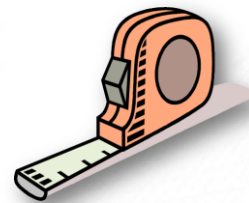
12/01/2026

19

19

Commonly Used Subjective Measurement Scales

- Likert-Type Scale
 - Evaluation-Type
 - Frequency-Type
 - Agreement-Type
- Semantic Differential Scale
- Summative Scale



12/01/2026

20

20

Likert Type Scales /1

■ Evaluation-type

■ Example:

- Familiarity with and comprehension of the software development environment (e.g., compiler, code generator, CASE tools):

- ☐ Little
- ☐ Unsatisfactory
- ☐ Satisfactory
- ☐ Excellent

12/01/2026

21

21

Likert Type Scales /2

■ Frequency-type

■ Example:

- Customers provided information to the project team
 - e.g., during interviews, when given questionnaires by the project staff, when presented with a “system walkthrough”, and/or when they are asked to provide feedback on a prototype:

- ☐ Never
- ☐ Rarely
- ☐ Occasionally
- ☐ Most of the time

12/01/2026

22

22

Likert Type Scales /3

■ Agreement-type

■ Example:

- The tasks supported by the software at the customer site were undertaking numerous changes during the project:

- ☐ Strongly Agree
- ☐ Agree
- ☐ Disagree
- ☐ Strongly Disagree

12/01/2026

23

23

Semantic Differential Scale

■ Items which include semantic opposites

■ Example:

- Processing of requests for changes to existing systems: the manner, method, and required time with which the MIS staff responds to user requests for changes in existing computer-based information systems or services.

Slow	□□□□□□	Fast
Timely	□□□□□□	Untimely

12/01/2026

24

24

Assigning Numbers to Scale Responses /1

- **Likert-Type Scales:**
- Ordinal Scale
- But: Often the distances between the four response categories are approximately (conceptually) equidistant [Spe76][Spe80] and thus are treated like approximate interval scales

☐ Strongly Agree	→ 1
☐ Agree	→ 2
☐ Disagree	→ 3
☐ Strongly Disagree	→ 4

12/01/2026

25

25

Assigning Numbers to Scale Responses /2

- **Semantic Differential Scale:**
- Ordinal scale, but again, often treated as interval scales

Slow	☐	☐	☐	☐	☐	☐	☐	Fast
	1	2	3	4	5	6	7	

12/01/2026

26

26

Assigning Numbers to Scale Responses /3

■ Summative Scale:

- A summative scale is constructed by forming an unweighted sum of n items:

$$Y = X_1 + X_2 + \dots + X_n$$

- This is commonly considered to be an approximately interval scale, but controversial
- If the X items are on a Likert-type scale then there is a good empirical basis for treating this as an approximately interval scale

12/01/2026

27

27

Exercise 1

Determine the best “scale” for each of the following measurements using each of the Nominal, Ordinal, Interval and Ratio scales only once.

- Measuring execution time of a program
- Classification of objects based on their color
- Measuring duration of various phases of projects (Project scheduling)
- Measuring complexity of many software modules by defining 4 complexity classes (Trivial, Simple, Moderate, Complex)

12/01/2026

28

28

Exercise 1 (cont'd)

- Software products categorized according to their compatibility with the operating system (i.e. Windows, Linux, MacOS, DOS, etc.).
- Internet services categorized according to their relevant technologies (i.e. dial-up, DSL, high-speed, wireless, etc.)
- Measuring attitudes towards an Internet service (say, on an n-point rating, $n = 0$ to 10).

12/01/2026

29

29

Exercise 1 (cont'd)

- Measuring attitude towards Internet services, if the evaluation scores are numerically meaningful so the difference between a rate of 3 and a rate of 6 is exactly the same as the difference between a rate of 7 and a rate of 10.
- Measuring attitude towards Internet services, if the differences between the data values are numerically meaningful and equal. In addition, a score of zero implies either the full absence of the service being evaluated or the full dissatisfaction with it.

12/01/2026

30

30

Exercise 2

- Suppose that you are asked to study various software development tools and recommend the best three to your company. The following table shows a list of available development tools.
- **Discuss with your partner (thảo luận nhóm 2)**

12/01/2026

31

31

Exercise 2 (cont'd)

Tool Name/Vendor	Languages Supported	Platforms	Features
Bean Machine IBM	Java	Windows, OS2, Unix	Best: Visual applet and JavaBean generation
CodeWarrior Pro Metrowerks	Java, C, C++, Pascal	Unix, Windows, Mac	Best: if you need to support Unix, Windows, and Mac platforms
Java Workshop Sun Microsystems	Java	Solaris, Windows	Better: Written 100% in Java; tools based on a web browser metaphor
JBuilder Imprise	Java	Windows, AS400	Better: database support
Visual Cafe for Java Symantec	Java	Windows	Good: multithreaded debugger
VisualAge IBM	Java	Unix, Windows	Good: includes incremental compiler and automatic version control
Visual J++ Microsoft	Java	Windows	Fair: All the bells and whistles for Windows

12/01/2026

32

32

Exercise 2 (cont'd)

- What are the *entities*, *attributes* and their *values* in your model?

<i>Entity</i>	<i>Attribute</i>	<i>Value</i>
Development Tool	Language supported	Java, C, C++, Pascal
	Platform	Win, Unix, Mac, OS2, AS400
	Feature	Fair, Good, Better, Best

12/01/2026

33

33

Are the following statements meaningful?

- Peter is twice as tall as Hermann
- Peter's temperature is 10% higher than Hermann's
- Defect X is more severe than defect Y
- Defect X is twice as severe as defect Y
- The cost for correcting defect X is twice as high as the cost for correcting defect Y
- The average temperature of city A (15 °C) is twice as high as the average temperature of city B (30 °C)
- Project Milestone 3 (end of coding) took ten times longer than Project Milestone 0 (project start)
- Coding took as long as requirements analysis

12/01/2026

35

35

Accuracy of Measurement /1

- In an ongoing project, it is possible to validate the estimate by comparing actual values with predicted values. If E is an estimate of a value and A is the actual value, the **relative error** (RE) in the estimate is:

$$RE = \frac{A - E}{A}$$

- Relative error will be negative or positive.
- Inaccuracies can result from either statistical variation or estimating bias.
- Estimating inaccuracies must be compensated for throughout the project.

12/01/2026

37

37

Accuracy of Measurement /2

- *Relative Error (RE)* can be calculated for a collection of estimates, e.g., it is useful to determine if predictions are accurate for a group of projects.
- The **mean relative error (MRE)** is calculated as the sum of RE for n projects divided by n . It is possible for small relative errors to balance out large ones, so the magnitude of the error can also be considered.
- The **mean magnitude of relative error (MMRE)** is calculated using the *absolute value* of RE .

12/01/2026

38

38

Accuracy of Measurement /3

- MMRE is used to define prediction quality.
- On a set of n projects, if k is the number of projects whose RE is less than or equal to q , then prediction quality is calculated as:

$$PRED(q) = k/n$$

- **Example:**

- $PRED(0.25) = 0.47$ means that 47% of the predicted values fall within 25% of their actual values.
- An estimation technique is (usually) acceptable if $PRED(0.25) = 0.75$.

12/01/2026

39

39

Exercise

- Assume you want to predict the error proneness of software modules.

a) How would you assess prediction accuracy?

$$RE \text{ (relative error)} = (A - E) / A$$

with A : actual, E : estimated

$$MRE = \frac{1}{n} \sum_{i=1}^n RE_i$$

$$MMRE = \frac{1}{n} \sum_{i=1}^n |RE_i|$$

b) How would you assess prediction quality?

$$PRED(q) = k/n$$

where k is the number of predictions with $|RE| \leq q$

12/01/2026

40

40

Example

- Setting failure intensity objective (λ) in terms of system availability (A):

$$A = \frac{1}{1 + \lambda t_m} \quad \text{or} \quad \lambda = \frac{1 - A}{A t_m}$$

λ is failure intensity (failure per unit of time)

t_m is downtime per failure

- The representation condition of the indirect measure of λ should preserve the relationship.

12/01/2026

87

41

Scale Invariance

- The scale of any indirect measure is deduced by means of the interrelation condition.

- **Example:** for a database system

$\lambda = 20$ /million transactions

$t_m = 6$ minutes

λ and t_m must have a compatible scale to calculate A

If there are 20,000 transactions/day and a day has 8 working hours, then t_m can be expressed as equivalent to 250 transactions and

$$A = 1 / (1 + \lambda t_m) = 1 / (1 + 0.005) = 0.995$$

12/01/2026

42

42

Discuss with your partner

- **Question : Determine which of the following statements are meaningful:**
 - a. Te length of Program A is 50.
 - b. Te length of Program A is 50 executable statements.
 - c. Program A took 3 months to write.
 - d. Program A is twice as long as Program B.
 - e. Program A is 50 lines longer than Program B.
 - f. Te cost of maintaining program A is twice that of maintaining Program B.
 - g. Program B is twice as maintainable as Program A.
 - h. Program A is more complex than Program B

12/01/2026

43

43

Assignment

- Question 1 Using measurement, what meaningful statements can we make about an attribute and the entities that possess it? For instance, is it meaningful to talk about doubling a design's quality? If not, how do we compare two different designs?

12/01/2026

44

44

Readings and Assignment 3

- You are required to read the paper in the links below and Answer the following questions. The answers will be submitted on e-Learning system
 - D. Pitts, "Why is software measurement hard?" [online] 1999. Available from <http://www.stickyminds.com>. Accessed Jan. 6, 2005
 - <https://www.youtube.com/watch?v=1B6qCBWsMYw>
- Question 1
 - What Are The Challenges Of Measurement?
 - Example OO Measurement?

12/01/2026

45

45

Software Measurement!



12/01/2026

46

46